| Civics Group | Index Number | Name (use BLOCK LETTERS) |
|---|---|---|
| | | |

**ST. ANDREW'S JUNIOR COLLEGE**
**2011 JC2 Preliminary Examinations**

**H2 BIOLOGY** **9648/1**

# Paper 1: Multiple Choice

Wednesday 21 September 2011 1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser (not supplied)
Soft pencil (type B or HB is recommended)

**READ THESE INSTRUCTIONS FIRST**

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on the multiple choice answer sheet in the spaces provided.

There are **40** questions in this paper. Answer all questions. For each question, there are four possible answers, A, B, C and D.

Choose the one you consider correct and record your choice in soft pencil on the separate multiple choice answer sheet.

**INFORMATION TO CANDIDATES**

Each correct answer will score one mark. A mark will not be deducted for wrong answer. Any rough working should be done in this booklet.

At the end of the examination, submit the multiple choice answer sheet only.

1 The correct shape of a globular protein is maintained by several types of bond.

Bond types 1 disulphide 3 ionic
2 hydrogen 4 peptide

Which bond types are involved in maintaining the different levels of structural complexity of a globular protein?

| | Level of structural complexity | | |
|---|---|---|---|
| | Primary | Secondary | Tertiary |
| **A** | 2 | 1 | 2, 3, 4 |
| **B** | 2 | 4 | 2, 3, 4 |
| **C** | 4 | 2 | 1, 2, 3 |
| **D** | 4 | 3 | 1, 2, 3 |

2 Three pure substances are analysed. The table shows the elements that each contains.

| | C | H | O | N | P | S |
|---|---|---|---|---|---|---|
| **X** | + | + | + | + | - | - |
| **Y** | + | + | + | - | - | - |
| **Z** | + | + | + | + | - | + |

Which combination could be correct?

| | X | Y | Z |
|---|---|---|---|
| **A** | adenine | carbohydrate | fat |
| **B** | adenine | fat | an amino acid |
| **C** | an amino acid | fat | carbohydrate |
| **D** | fat | carbohydrate | an amino acid |

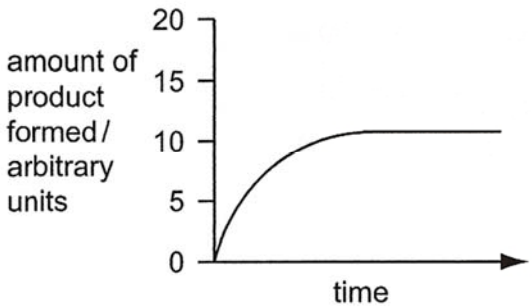
3 The list shows three characteristics of enzyme activity.

1 Reaction rate decreases if the concentration of non-competitive inhibitor increases.
2 Reaction rate is reduced at extremes of pH.
3 Reaction rate is reduced at low temperature.

What explains each of these characteristics?

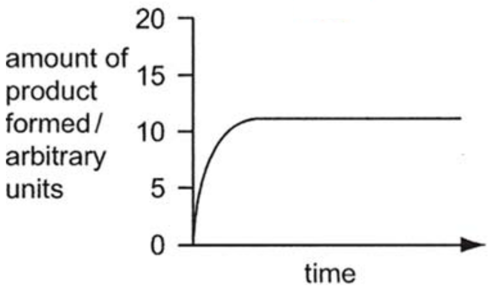
| | Availability of active site is reduced. | Reduced kinetic energy reduces the rate of molecular collisions. | Hydrogen bonding is disrupted. |
|---|---|---|---|
| **A** | 1 | 2 | 3 |
| **B** | 2 | 1 | 3 |
| **C** | 1 | 3 | 2 |
| **D** | 2 | 3 | 1 |

**4** The graph shows the amount of product formed by a standard concentration of enzyme and a standard concentration of substrate at a temperature of 15°C.

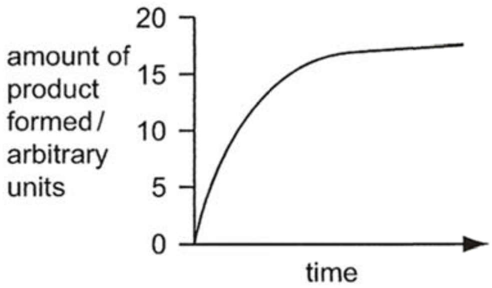



Which graph shows the effect on the activity of the enzyme of increasing the temperature to 20°C?

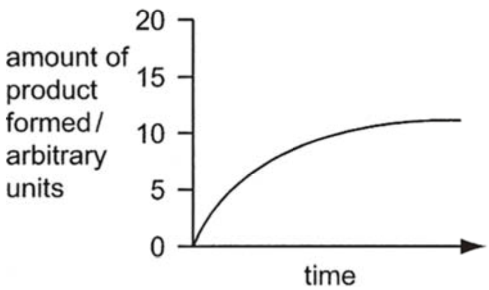
**A**



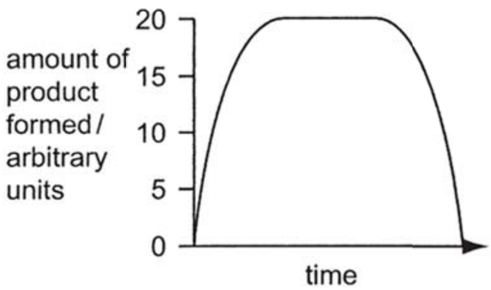

**B**




**C**



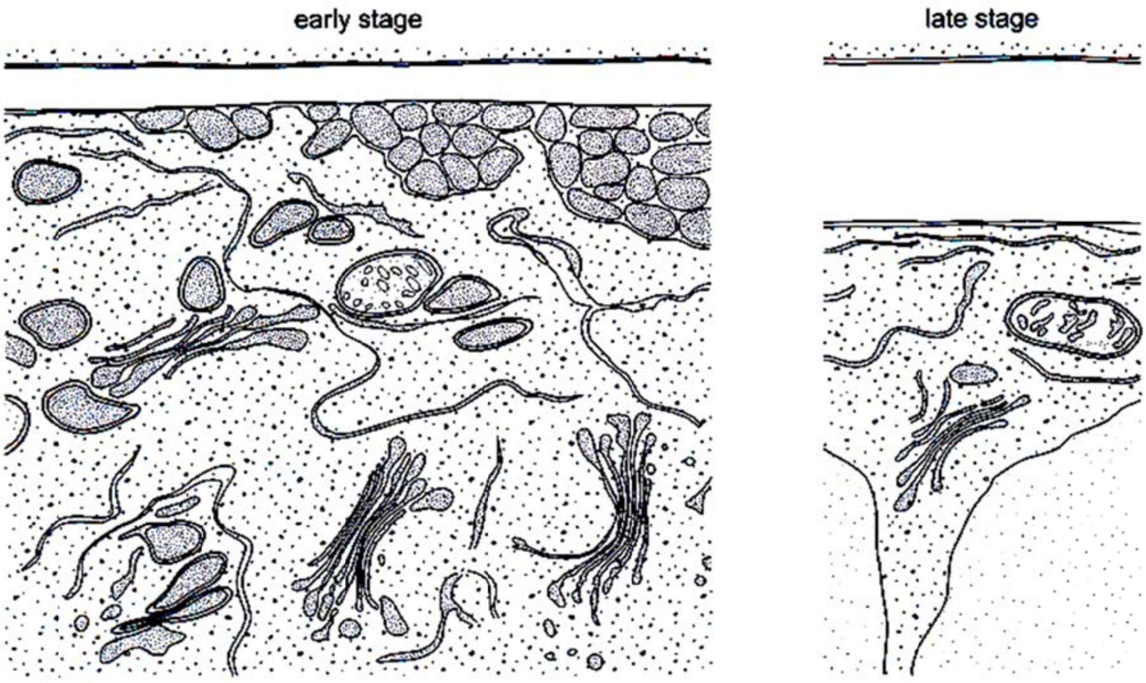

**D**




**5** A severe inherited condition arises from the failure to produce an enzyme that breaks down glycoproteins in cells. The condition can be diagnosed from an electron micrograph of a patient’s cells. Which abnormality would be observed in these cells?

**A** An incomplete chromosome due to the lack of a gene.
**B** Larger lysosomes due to the accumulation of glycoproteins.
**C** Thinner cell surface membrane due to the lack of glycoproteins.
**D** Less endoplasmic reticulum due to a reduction in protein synthesis.

**6** The photo electron micrographs show early and late stages in the development of the cell wall in a young plant cell.




Which statement describes the events leading to the development of the cell wall?

**A** Complex carbohydrates assembled in the Golgi body are exported to the cell wall by the Golgi vesicles.
**B** Enzymes in the cell surface membrane synthesise the cell wall components from soluble carbohydrates brought by the Golgi vesicles.
**C** Polysaccharides are exported to the cell wall and synthesized into wall components by the Golgi body.
**D** Ribosomes synthesise glycoproteins that are exported by Golgi vesicles to be used in the cell wall.

**7** A certain cell surface membrane is made up entirely of phospholipids and is 4 nm thick. A volume of 1 $mm^3$ of this membrane was homogenised and dropped onto a large tray of water. The phospholipids spread out to form a continuous thin film.

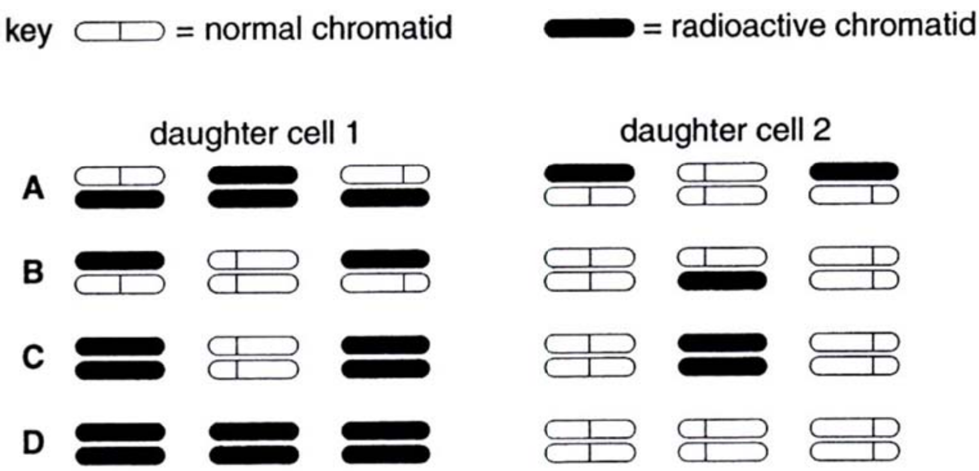
What is the expected area of this film?

**A** 250 000 $mm^2$ because the molecules originally exist as a single layer
**B** 250 000 $mm^2$ because the molecules originally exist as a double layer
**C** 500 000 $mm^2$ because the molecules originally exist as a single layer
**D** 500 000 $mm^2$ because the molecules originally exist as a double layer

**8** The following experiment was carried out.

1 Haploid cells, containing three chromosomes each, were grown in a medium containing radioactive thymine, so that all the DNA was labelled.
2 Cells in early interphase were then transferred to a medium where the available thymine was not radioactive.
3 A single cell was immediately isolated and allowed to divide once. When the two daughter cells reached the next metaphase, they were fixed and their three chromosomes were inspected for radioactivity.

Which diagram represents the distribution of radioactivity at metaphase in the two daughter cells?




**9** How do viruses cause diseases in animals?

1 They degrade the host cell's chromosomes.
2 They disrupt the oncogenes of the host cell causing uncontrolled cell division.
3 Their viral proteins and glycoproteins on the surface membrane of host cells cause them to be recognized and destroyed by the body's immune defenses.

**A** 1 and 2
**B** 1 and 3
**C** 2 and 3
**D** All of the above

**10** Bacterial Strain A has been infected with viruses. Upon lysis, the virions produced are introduced to Bacterial Strain B. After a short period of time, a new strain of bacteria is detected that is very similar to Strain A but has a few characteristics of Strain B.

Which is the process that leads to the production of the new bacterial strain?

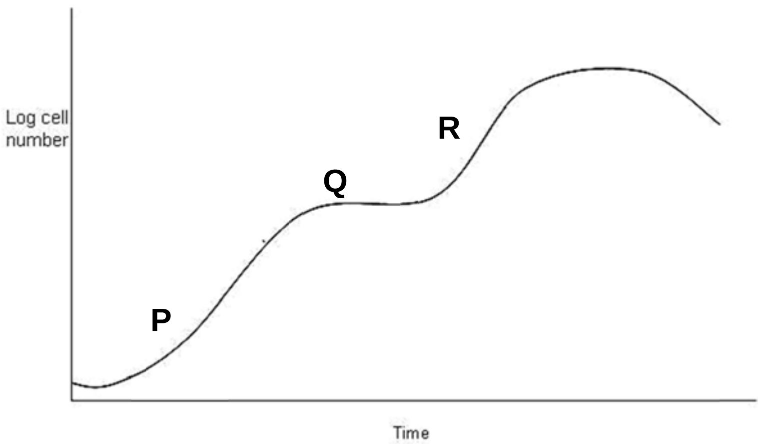
**A** genetic drift
**B** transduction
**C** transformation
**D** conjugation

**11** When a bacterial cell is infected by a lambda phage, this may not result in the immediate death of the bacterial cell. Which of the following statements is/are possible mechanism/s for the phenomenon?

1 The transcription and translation of viral DNA is not initiated.
2 A repressor protein is synthesized to shut down replication of the phage.
3 Mutation of host genome results in the inability of lambda phage to enter lytic cycle.

**A** 1 only
**B** 1 and 2
**C** 2 and 3
**D** 1, 2 and 3

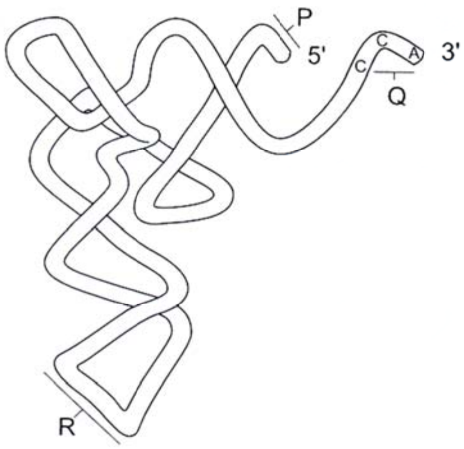
**12** *E.coli* bacteria are grown in a culture of nutrients which includes glucose and lactose as the main source of carbon-based nutrient.




Which of the following statements accounts for the growth curve showing the reproduction of *E.coli* bacteria?

| | P | Q | R |
|---|---|---|---|
| **A** | Lactose is metabolised | Production of enzymes for lactose metabolism is repressed | Glucose is metabolised |
| **B** | Lactose is metabolised | Enzymes for lactose metabolism is produced | Glucose is metabolized |
| **C** | Glucose is metabolised | Production of enzymes for lactose metabolism is repressed | Lactose is metabolised |
| **D** | Glucose is metabolised | Enzymes for lactose metabolism is produced | Lactose is metabolised |

**13** The diagram shows a tRNA molecule.



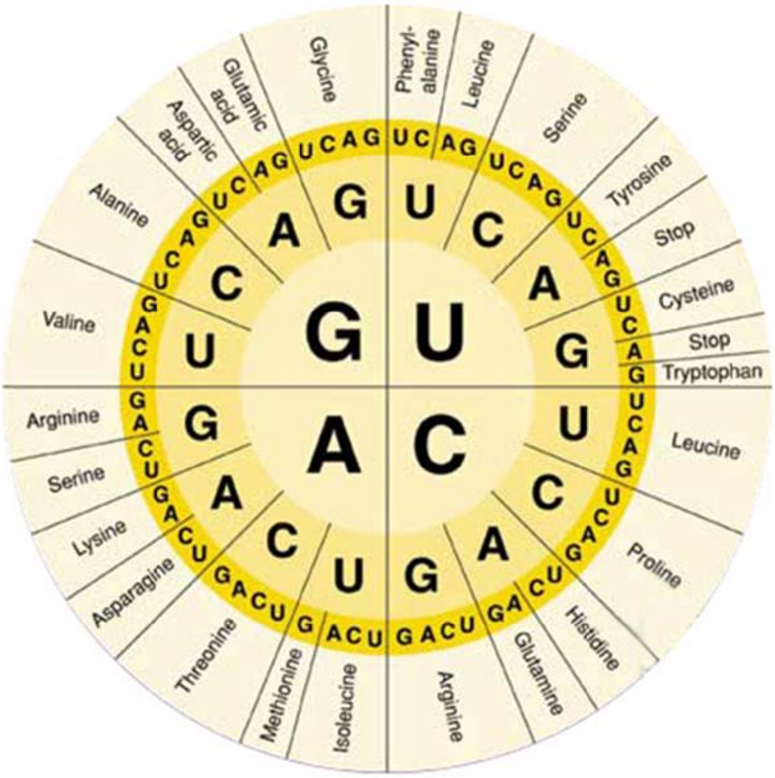

What are the roles of the parts of the molecule labelled P, Q and R?

| | Binding of mRNA | Binding of amino acids | Binding of peptide |
|---|---|---|---|
| **A** | P | Q | R |
| **B** | P | R | Q |
| **C** | Q | P | R |
| **D** | R | Q | P |

**14** A pentapeptide has the following amino acid sequence :

$H_2N$ – Met – Leu – Trp – Ala – Phe – COOH

Where Met : Methionine Ala : Alanine
Leu : Leucine Phe : Phenylalanine
Trp : Tryptophan




Using the genetic code shown in the diagram above, deduce the corresponding template strand sequence during transcription.

**A** 5' – GAA GGC CCA CAA CAT – 3'
**B** 5' – TTA GAA GGC CCA CAA CAT – 3'
**C** 5' – ATG TTG TGG GCC TTT TAG – 3'
**D** 5' – AUG CUG UGG GCA UUC – 3'

**15** A geneticist introduced a transgene into yeast cells and isolates five independent cell lines in which the transgene has integrated into the yeast genome. In four of the lines, the transgene is expressed strongly, but in the fifth there is no expression at all. A likely explanation for the lack of transgene expression in the fifth cell line is that the

**A** Transgene was integrated into a euchromatic region of the genome.
**B** Transgene was integrated into a heterochromatic region of the genome.
**C** Transgene was mutated during the process of integration into the host cell genome.
**D** Transgene was integrated into a region of the genome characterized by high histone acetylation.

**16** Which of the following mutations would lead to the occurrence of cancer?

1 germline mutation in a copy of p53 gene
2 somatic mutation in second copy of p53 gene
3 gene amplification of Ras gene
4 point mutation in a copy of Ras gene

**A** 1 and 4
**B** 2 and 3
**C** 2, 3 and 4
**D** 1, 2, 3 and 4

**17** Which of the following statement is true of the end-replication problem and telomerase?

1 Telomerase has reverse transcriptase activity.
2 Telomerase prevents the end-replication problem from occurring.
3 In germ-line cells, telomeric repeat sequences are added to the 3' end of newly synthesized daughter strands.

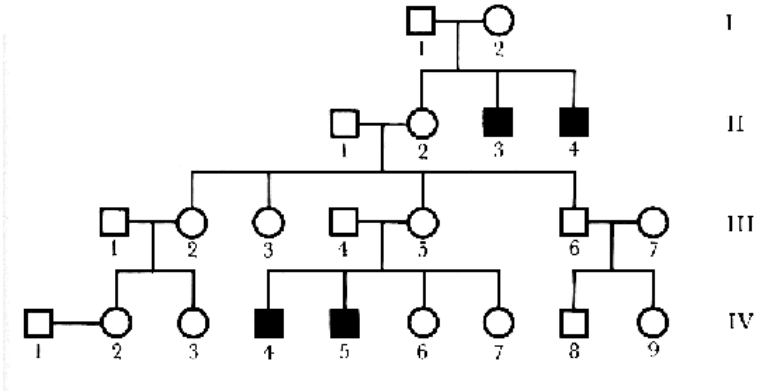
**A** 1 only
**B** 1 and 2
**C** 2 and 3
**D** 1, 2 and 3

**18** Down's syndrome can be caused by two forms of chromosome aberration. Classic Down's syndrome is a trisomy of chromosome 21 that affects all cells. A mosaic condition occurs when there are two or more cell types in the body with differences in chromosome number and structure.

Suggest how the mosaic condition arises?

**A** non-disjunction of chromosomes in mitosis in early fetal development
**B** non-disjunction of chromosomes in meiosis during formation of the ovum
**C** non-disjunction of chromosomes in meiosis during formation of sperm
**D** translocation of chromosomes at maturation of the ovum

**19** Haemophilia is a rare genetic disorder in which the blood does not clot normally. The mutation that causes haemophilia is located on the X-chromosome. The pedigree below shows the inheritance of haemophilia in a family. Squares represent males while circles represent females. Dark symbols represent affected individuals.




What is the probability that a son born to IV-2 will be a haemophiliac?

**A** 0.5
**B** 0.25
**C** 0.125
**D** 0.0625

**20** Fruit flies *Drosophila* homozygous for long wings, were crossed with flies homozygous for vestigial wings. The $F_1$ and $F_2$ generations were raised at three different temperatures.

At each temperature, the $F_1$ generation all had long wings.

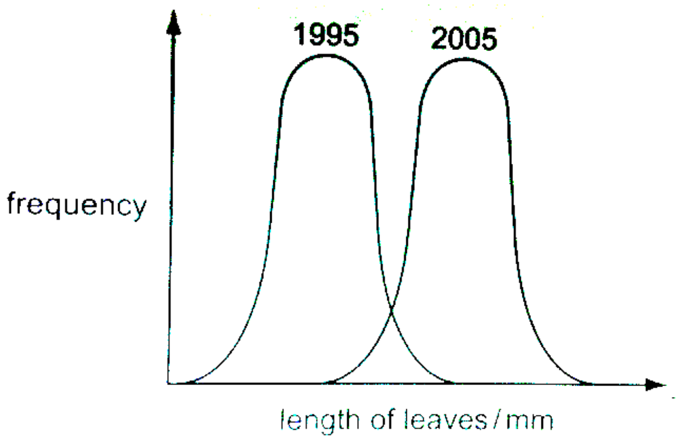
The table below shows the results in the $F_2$ generation.

| ***Temperature*** | ***Result*** |
|---|---|
| 21°C | ¾ long wings, ¼ vestigial wings |
| 26°C | ¾ long wings, ¼ intermediate wing length |
| 31°C | all long wings |

Which statement explains these results?

**A** Wing length is under polygenic control.
**B** Long wing and vestigial wing illustrate codominance at 26°C.
**C** Heterozygous flies have vestigial wings only at 21°C or below but have long wings at 31°C or above.
**D** Vestigial wing is recessive but causes a vestigial wing phenotype only at lower temperatures.

**21** The graph shows the results of data collected by measuring the leaf length of the same hybrid of coffee plants grown in 1995 and 2005.




What explains the difference in leaf length of the coffee plants?

**A** continuous variation and a genetic factor
**B** continuous variation and an environmental factor
**C** discontinuous variation and a genetic factor
**D** discontinuous variation and an environmental factor

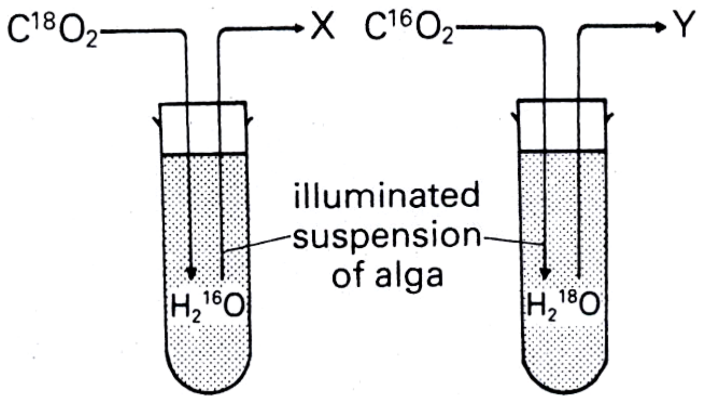
**22** The table shows the amount of carbon dioxide taken up by a plant in the light and the amount of carbon dioxide produced in the dark, at different temperatures.

| **Temperature / °C** | **Net uptake of $CO_2$ in light / mg $g^{-1}h^{-1}$** | **Release of $CO_2$ in dark / mg $g^{-1}h^{-1}$)** |
|---|---|---|
| **5** | 1.1 | 0.2 |
| **15** | 2.8 | 0.8 |
| **30** | 2.0 | 2.4 |

Assuming that the rate of respiration of this plant is the same in light and dark, which of the following statement is correct.

**A** The true uptake of carbon dioxide in the light at 15°C is 2.0 mg $g^{-1}h^{-1}$.
**B** Temperature has no effect on the rate of carbon dioxide uptake in the light.
**C** The rate of photosynthesis exceeds the rate of respiration at all temperatures.
**D** None of the above

**23** In the experiment shown in the diagram, the oxygen given off at **X** and **Y** will be labelled with the isotopes _____ and _____ respectively.



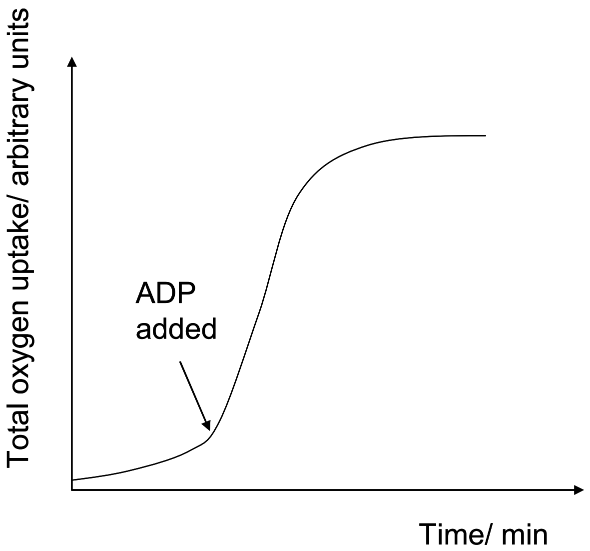

| | X | Y |
|---|---|---|
| **A** | $^{16}O$ | $^{16}O$ |
| **B** | $^{16}O$ | $^{18}O$ |
| **C** | $^{18}O$ | $^{16}O$ |
| **D** | $^{18}O$ | $^{18}O$ |

**24** 2,4-dinitrophenol is a chemical that targets the inner mitochondrial membrane to uncouple oxidative phosphorylation from electron transport. It allows protons to cross the inner mitochondrial membrane and thus dissipates the proton gradient.

Which of the following is true of a cell treated with 2,4-dinitrophenol?

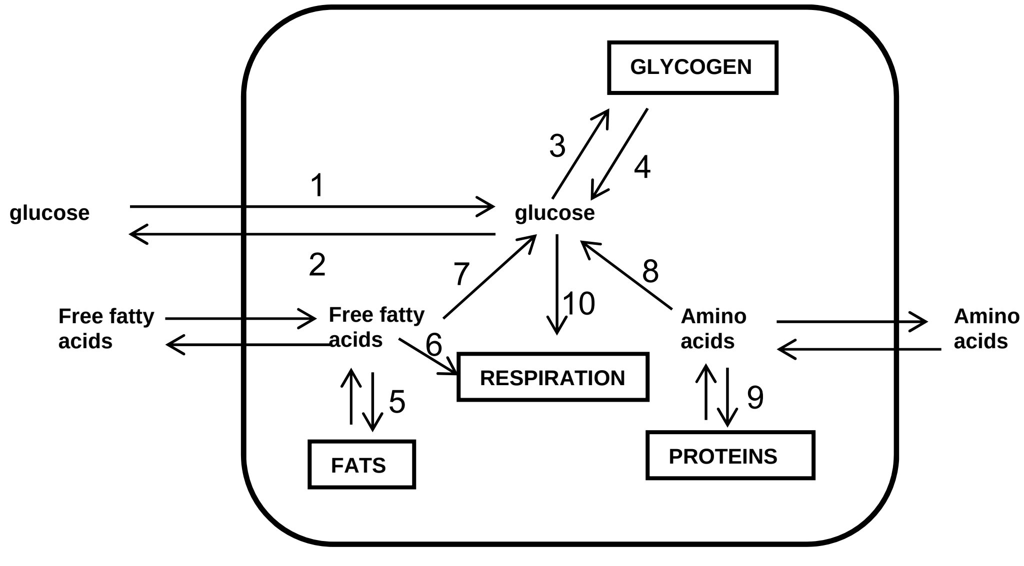
| | **Ability to use glucose** | **Ability to use oxygen** | **Ability to produce carbon dioxide** |
|---|---|---|---|
| **A** | Yes | Yes | Yes |
| **B** | Yes | No | Yes |
| **C** | Yes | No | No |
| **D** | No | No | No |

**25** The graph shows the total oxygen uptake by a sample of homogenised liver tissue over a period of time. Which statement explains the sharp rise in oxygen uptake after the addition of ADP?




**A** ADP reduces the molecular oxygen to water.
**B** Electron transport results in the oxidation of ADP.
**C** Electron transport requires the hydrolysis of ADP.
**D** Electron transport is accompanied by simultaneous phosphorylation of ADP.

**26** The diagram below shows some biochemical pathways in a liver cell. During which processes will the hormone glucagon exert an effect?




**A** (1), (3), (5), (9)
**B** (1), (4), (8), (10)
**C** (2), (7), (9), (10)
**D** (4), (6), (7), (8)

**27** In a clinical trial, the effectiveness of two types of synthetic drug for a certain disease was tested. Participants in this test were divided into two groups. One group received Drug A. The second group received Drug B. All participants received the same amount and concentration of the appropriate drug.

The table below shows the average blood glucose levels after injection of the drug for both groups of participants **after a standardised meal**. Normal blood glucose level for a healthy person is 90mg/100ml.

| Time after injection (h) | Average blood glucose level in participants injected with **Drug A** (mg/100ml) | Average blood glucose level in participants injected with **Drug B** (mg/100ml) |
|---|---|---|
| 0 | 360 | 365 |
| 1 | 231 | 326 |
| 2 | 180 | 271 |
| 3 | 160 | 222 |
| 4 | 160 | 152 |
| 5 | 160 | 120 |
| 6 | 160 | 90 |

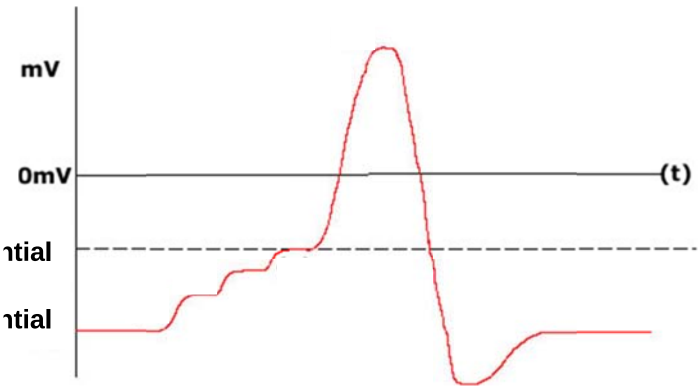
Which of the following statements is **not** true?

**A** Drug A is useful for type II diabetes.
**B** Drug A may be degraded roughly 3 hours after injection.
**C** Drug A must be injected more frequently than drug B in a day.
**D** Drug B resembles a hormone released from the beta cells of islet of Langerhans of pancreas.

**28** A neurone is undergoing relative refractory period. Which of the following statements is true?

**A** Membrane potential is less negative than resting potential.
**B** There is delayed closing of voltage-gated potassium channels.
**C** It is possible to initiate another action potential if a weaker stimulus is applied.
**D** It is impossible to initiate another action potential regardless of the stimulus strength.

**29** The diagram below shows the membrane potential of a post-synaptic neurone. Which of the following statements are likely to correspond to the diagram?




1 repeated release of neurotransmitter from one pre-synaptic terminal before the post-synaptic potential returns to its resting potential
2 release of neurotransmitters from several different pre-synaptic terminals simultaneously
3 chloride ion channels on post-synaptic membrane are opened
4 sodium ion channels on post-synaptic membrane are opened

**A** 1 and 3
**B** 1 and 4
**C** 2 and 3
**D** 2 and 4

**30** In an example of cell signalling, testosterone binds to its receptor, causing a change in conformation of the receptor (dimerisation). The dimerised receptor then attach to specific DNA sequences directly to regulate transcription of certain genes.

Which of the following statement about this cell signalling pathway is true?

**A** There is a phosphorylation cascade involved.
**B** Receptors are found on the cell surface membrane.
**C** Receptors for testosterone may be transcription factors.
**D** Secondary messengers are involved in signal transduction.

**31** A scientist is investigating the effects of Poison T on the cell signaling pathway of glucagon. It is found that Poison T diminishes the effect of glucagon.

The table below shows the different components of the cell signaling pathway of glucagon and their statuses.

| **Component** | **Status** |
|---|---|
| G Protein Coupled Receptor (GPCR) | Configuration can be changed |
| G protein | Can exchange GDP for GTP |
| cAMP levels | low |
| Protein Kinase A | Inactivated |

Which of the following statements can be concluded from the results?

1 Poison T interferes with reception.
2 Poison T prevents G protein from hydrolysing GTP.
3 Poison T inactivates the enzyme adenylate cyclase.
4 Poison T stops the relaying of message down the phosphorylation cascade.

**A** 1 and 2
**B** 2 and 3
**C** 2 and 4
**D** 3 and 4

**32** A newly discovered hormone has been found to act on Receptor Tyrosine Kinases (RTK). Receptors exist as individual units originally.

Arrange the following steps of cell signalling in order, from first to last.

1 Auto-crossphosphorylation.
2 Activation of relay proteins.
3 Tyrosine kinase domains of each receptor molecule activated.
4 Ligand binds to RTK.
5 Recognition of specific relay proteins.
6 Dimerization of two receptor molecules.

**A** 4, 1, 6, 3, 5, 2
**B** 4, 5, 2, 6, 3, 1
**C** 4, 6, 3, 1, 5, 2
**D** 5, 2, 4, 6, 1, 3

**33** Which of the following descriptions refers accurately to homologous structures?

| | **Basic structure** | **Common ancestor** | **Function performed** |
|---|---|---|---|
| **A** | Same | No | Different |
| **B** | Different | No | Same |
| **C** | Same | Yes | Different |
| **D** | Different | Yes | Same |

**34** Which sequence of events correctly describes evolution?

1 Differential reproduction of the spiders occurs.
2 A new selection pressure occurs.
3 Allele frequencies within the spider population change.
4 Poorly adapted spiders have decreased survivorship.

**A** 2 4 1 3
**B** 2 4 3 1
**C** 4 1 3 2
**D** 4 3 1 2

**35** A large population of equal numbers of dark and light mice was released into an area where owls are predators of mice. Because of predation, only 25 % of these mice survived and selective killing by owls changed the proportions of dark to light mice from 1:1 to 4:1.

If mice produce an average of eight offspring per litter and the pattern of predation remains the same, what would happen to the population of light mice?

**A** They disappear after one further generation.
**B** They disappear after two further generations.
**C** They remain at the level of one fifth of the population.
**D** They will be reduced to a constant but very low frequency.

**36** Which of the following statement about a genomic library and cDNA library is true?

**A** The genomic library contains only the genes that can be expressed in the cell.
**B** A genomic library varies, dependent on the cell type used to make it, whereas the content of a cDNA library does not.
**C** A genomic library contains only noncoding sequences, whereas a cDNA library contains only coding sequences.
**D** A genomic library can be made using a restriction enzyme and DNA ligase only, whereas a cDNA library requires both of these as well as reverse transcriptase and DNA polymerase.

**37** Stem cells are found in many tissues that require frequent cell replacement such as the skin, the intestine or the blood.

However, within their own environments, a bone marrow cell cannot be induced to produce a skin cell and a skin cell cannot be induced to produce a bone marrow cell. Which statement explains this?

**A** Genes not required for a particular cell line are methylated.
**B** mRNA that is not required for a particular cell line is destroyed.
**C** Genes not required for a particular cell line are removed using restriction enzymes.
**D** Different stem cells have only the genes required for their particular cell line.

**38** Non-viral *ex vivo* transfer of a gene encoding coagulation factor VII was performed using fibroblast cells isolated from patients suffering from severe haemophilia A.

What is the sequence of events for the *ex vivo* transfer of the gene encoding coagulation factor VIII?

1 Transfection with plasmids containing the gene encoding coagulation factor VII
2 Implantation of cells into patients
3 Isolation of fibroblasts
4 Selection and cloning of cells expressing coagulation factor VII

**A** 2, 3, 1, 4
**B** 3, 1, 4, 2
**C** 3, 2, 4, 1
**D** 3, 4, 2, 1

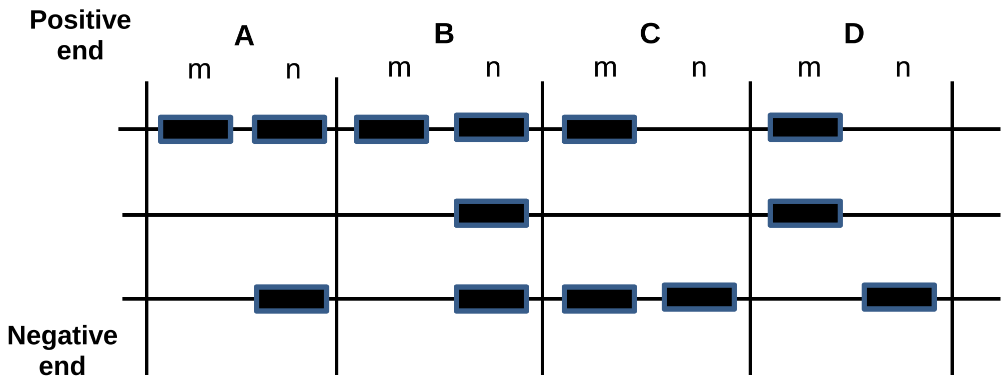
**39** The basis for development of leukaemia in some children treated with a retrovirus-derived gene therapy vector is:

**A** Use of a virus that contains an oncogene
**B** Insertion of the viral genome adjacent to an oncogene
**C** Insertion causing a mutation of a tumour suppressor gene
**D** Stimulation of immune response to cell surface antigens introduced by an inserted gene

**40** There are two forms of the seeds of the garden pea, *Pisum sativum*, one with a wrinkled skin and the other with a smooth skin. Plants with wrinkled seeds have a recessive mutation caused by the insertion of a few nucleotides into a gene.

Two heterozygous pea plants were crossed. Two of the offspring (m and n) had the DNA for this gene removed and separated by electrophoresis.

Which pattern of bands shows a wrinkled phenotype and a smooth phenotype?

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